Week 1 Assignment

Setup and first steps in R

Complete this after Wednesday's tutorial. As with all assignments, this is graded on completion, not correctness — the point is to show you engaged with the material. Once completed, render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 2's lecture (Monday). Late assignments follow the policy in the syllabus: up to three weeks late, docked 2% per day.

Part 1: Reflection

In 3–4 sentences, answer: why might a biologist care about *designing* an experiment carefully, rather than just collecting data and running a statistical test afterward? Give a concrete example if you can, from your own subfield or otherwise.

Part 2: R warm-up

1. Create a vector of 8 numeric values representing a measurement of your choosing (e.g., body lengths, leaf counts, reaction times; incorporate the trait type into the object name).
2. Compute its mean, standard deviation, minimum, and maximum using built-in R functions — don't hardcode the answers.
3. Make a histogram of your vector using `ggplot2`.
4. In one sentence, describe the shape of the distribution you see (symmetric? skewed? any obvious outliers?).